

32.1

Guass Seidal Iterative Method (1)

32.1.1 Guass Seidal Iterative Method: ISRO-2013-48



The Guass-Seidal iterative method can be used to solve which of the following sets?

- A. Linear algebraic equations
- B. Linear and non-linear algebraic equations
- C. Linear differential equations
- D. Linear and non-linear differential equations

isro2013 numerical-methods guass-seidal-iterative-method

Answer key

32.2

Interpolation (1)

32.2.1 Interpolation: ISRO CSE 2011 | Question: 52



Given

X: 0 10 16

Y: 6 16 28

The interpolated value $X=4$ using piecewise linear interpolation is

- A. 11
- B. 4
- C. 22
- D. 10

isro2011 interpolation non-gatecse

Answer key

32.3

Newton Raphson (1)

32.3.1 Newton Raphson: ISRO CSE 2017 | Question: 3



Using Newton-Raphson method, a root correct to 3 decimal places of $x^3 - 3x - 5 = 0$

- A. 2.222
- B. 2.275
- C. 2.279
- D. None of the above

isro2017 newton-raphson non-gatecse

Answer key

32.4

Numerical Methods (5)

32.4.1 Numerical Methods: GATE CSE 2006 | Question: 1, ISRO2009-57



Consider the polynomial $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$, where $a_i \neq 0, \forall i$. The minimum number of multiplications needed to evaluate p on an input x is:

- A. 3
- B. 4
- C. 6
- D. 9

gatecse-2006 numerical-methods normal isro2009

Answer key

32.4.2 Numerical Methods: ISRO CSE 2009 | Question: 44



A root α of equation $f(x) = 0$ can be computed to any degree of accuracy if a 'good' initial approximation x_0 is chosen for which

- A. $f(x_0) > 0$
- B. $f(x_0)f''(x_0) > 0$
- C. $f(x_0)f''(x_0) < 0$
- D. $f''(x_0) > 0$

Answer key 

32.4.3 Numerical Methods: ISRO CSE 2009 | Question: 46



The shift operator E is defined as $E[f(x_i)] = f(x_i + h)$ and $E'[f(x_i)] = f(x_i - h)$ then Δ (forward difference) in terms of E is

- A. $E - 1$ B. E C. $1 - E^{-1}$ D. $1 - E$

Answer key 

32.4.4 Numerical Methods: ISRO CSE 2009 | Question: 47



The formula $\int_{x_0}^{x_n} y(n) dx \simeq h/2(y_0 + 2y_1 + \dots + 2y_{n-1} + y_n) - h/12(\nabla y_n - \Delta y_0) - h/24(\nabla^2 y_n + \Delta^2 y_0) - 19h/720(\nabla^3 y_n - \Delta^3 y_0) \dots$ is called

- A. Simpson rule B. Trapezoidal rule
C. Romberg's rule D. Gregory's formula

Answer key 

32.4.5 Numerical Methods: ISRO CSE 2009 | Question: 51



The formula $P_k = y_0 + k\nabla y_0 + \frac{k(k+1)}{2}\nabla^2 y_0 + \dots + \frac{k \dots (k+n-1)}{n!}\nabla^n y_0$ is

- A. Newton's backward formula B. Gauss forward formula
C. Gauss backward formula D. Stirling's formula

Answer key 

32.5 Polynomials (1)

32.5.1 Polynomials: ISRO CSE 2009 | Question: 48



The cubic polynomial $y(x)$ which takes the following values: $y(0) = 1, y(1) = 0, y(2) = 1$ and $y(3) = 10$ is

- A. $x^3 + 2x^2 + 1$
B. $x^3 + 3x^2 - 1$
C. $x^3 + 1$
D. $x^3 - 2x^2 + 1$

Answer key 

Answer Keys

32.1.1	A	32.2.1	D	32.3.1	C	32.4.1	A	32.4.2	A
32.4.3	A	32.4.4	B	32.4.5	A	32.5.1	D		